

STA2503: MATHEMATICAL FINANCE REPORT TEMPLATE

A. AUTHOR, B. AUTHOR, AND C. AUTHOR

ABSTRACT. Here is where you would put your executive summary of the project – no more than 2 short paragraphs.

1. INTRODUCTION

Here is where you would put an introduction about what you are doing and what are your main findings.

2. METHODOLOGY

Here is where you would describe more details about the model and methodology
Here are some sample equations...

$$(1) \quad \varphi_{\tau}^{\nu} = \underbrace{\mathbf{n}(A_{\tau} - S_0)}_{\text{TWAP spread}} - \left(\underbrace{\int_0^{\tau} \nu_u \hat{S}_u^{\nu} du}_{\text{cost of cotinuous trading}} + \underbrace{q_{\tau-} (S_{\tau}^{\nu} + \alpha(S_{\tau}^{\nu}) q_{\tau-})}_{\text{cost of block trade}} \right).$$

$$(2a) \quad C = \{ (t, S, A, q) : H(t, S, A, q) > \mathbf{n}(A - S_0) - \ell(q, S) \} \quad \text{continuation region,}$$

$$(2b) \quad D = \{ (t, S, A, q) : H(t, S, A, q) \leq \mathbf{n}(A - S_0) - \ell(q, S) \} \quad \text{stopping region,}$$

$$(2c) \quad E = \{ (t, S, A, q) : H(t, S, A, q) = \mathbf{n}(A - S_0) - \ell(q, S) \} \quad \text{exercise boundary.}$$

You can reference the equations individually: C is in (2a), D is in (2b), E is in (2c) and all together they are CDE are all in (2).

You can include .pdf or .png figures as in Figures 1 and 2 below. Notice that I have used large enough fonts on axis titles and numbering so that everything is easy to read.

You can generate .pdf from matlab figures by using the `save2pdf.m` function. When a figure is selected, calling `save2pdf('AFigure')` will save a .pdf with name **AFigure** in the current working directory exactly as it appears on the screen.

3. RESULTS

Here is where you would put results and any financial intuition.

3.1. **A subsection.** Below, Table 1 shows a simple example table.

3.2. **Another Subsection.** Table 2 shows a more complex table.

4. CONCLUSIONS

Write some concluding remarks here.

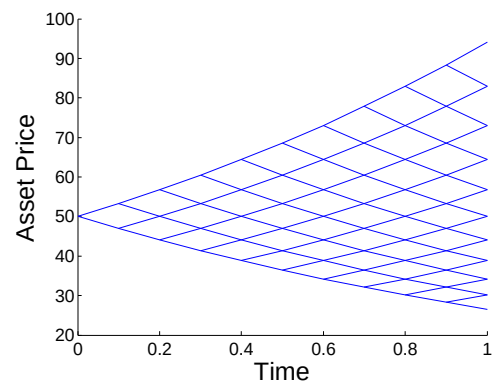


FIGURE 1. A binomial tree.

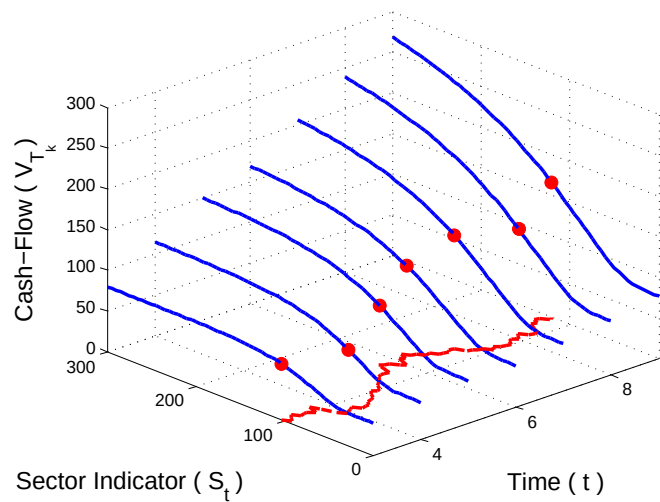


FIGURE 2. This diagram shows a sample path of the sector indicator and its transformation to cash-flows.

Team	P	W	D	L	F	A	Pts
Manchester United	6	4	0	2	10	5	12
Celtic	6	3	0	3	8	9	9
Benfica	6	2	1	3	7	8	7
FC Copenhagen	6	2	1	3	5	8	7

TABLE 1. A very simple table.

Case I: $\varepsilon = 0.04$, $\rho = 0.7$ and liquidation costs = 1bp					
ϕ	α_t		Bench.	$\rho = 1$	$\rho = 0$
1×10^{-5}	Yes	mean	14.16	12.97	9.96
		(std.)	(6.84)	(7.31)	(5.88)
	No	mean	-3.82	-3.86	-4.35
		(std.)	(2.85)	(2.85)	(2.99)
2×10^{-5}	Yes	mean	13.42	12.36	9.68
		(std.)	(5.35)	(5.64)	(4.75)
	No	mean	-1.68	-1.79	-2.82
		(std.)	(2.09)	(2.08)	(2.19)
4×10^{-5}	Yes	mean	12.27	11.34	9.19
		(std.)	(4.23)	(4.41)	(3.86)
	No	mean	0.07	-0.06	-1.27
		(std.)	(1.50)	(1.49)	(1.60)

TABLE 2. The results from some simulation.